

Thomas J. Johnston

ENGINEERING LEADER

thomasjohnstonjames@gmail.com

San Francisco, CA

linkedin.com/in/johnstonthomas

EDUCATION

B.S. Computer Science

University of Illinois Urbana-Champaign · 2010 – 2014

SKILLS

APPLIED AI

AI-Assisted Development
LLM Applications AI Agents
Conversational AI / Voice UI
Entity Resolution

DISTRIBUTED SYSTEMS & ARCHITECTURE

Systems Design AWS
API Design Scalability
High Availability
Database Design

ENGINEERING EXCELLENCE

CI/CD Integration Testing
Deployment Pipelines

STATISTICS & EXPERIMENTATION

Experiment Design
A/B Testing
Statistical Inference
Latency & Metrics Analysis

PROGRAMMING LANGUAGES

Java Python JavaScript
C/C++ HTML/CSS

ENGINEERING MANAGEMENT

Hiring Mentorship Agile
Sprint Planning
Roadmap Design
Performance Management
OKRs/KPIs

Senior software engineer and engineering manager with 12+ years of industry experience, including 10 years at Amazon.com. Shipped IMDb into the original Alexa Knowledge Base, launched Alexa experiences internationally, delivered Voice Support for Thursday Night Football, and as a Software Development Manager grew a team from 5 to 10. Now an AI-native builder looking to lead again.

EXPERIENCE

Amazon.com

Jan 2016 – Jan 2026

Senior Software Engineer

May 2025 – Jan 2026

- Built a new Games search vertical for Fire TV end-to-end, catalog indexing, ranking, and presentation. Delivered the V1 in half the expected time using AI-assisted development to champion AI use on the team.
- Authored the architecture proposal to consolidate legacy voice services into the broader search architecture, driving alignment with director-level leadership.
- Founded and curated a weekly design review across the search organization.

Software Development Manager III

Oct 2022 – May 2025

- Shipped Voice Support for Thursday Night Football, Alexa's first Live Events use-case with millions of interactions per game. Crossed organizational boundaries to solve problems during the games in real-time.
- Grew a team of 5 engineers to 10, mentoring engineers from college hires to established professionals.
- Shipped rearchitecture of FireTV's Search Backend to support live, linear and sports search, a 10x increase in the scale of the catalog and search indexes.
- When merging two organizations of 25+ total teams and facing redundancies between team charters, built a technical architecture consensus with Principal Engineers and Leaders to establish a mandate and 3 year vision for the team.
- Promoted two engineers to Senior Engineer. Managed two summer internships.
- Calibrated Interviewer with over 30 interviews for roles across engineering levels including Engineer, Senior Engineer, and Senior Technical Program Manager roles.

Senior Software Engineer

Jan 2016 – Oct 2022

- Shipped IMDb knowledge ingestion into Alexa's original knowledge base, resolving entity collisions across the existing graph and supporting new question shapes without regressing Q&A accuracy, in partnership with ML scientists and knowledge engineers.
- Launched Alexa's voice experience in UK and DE as part of Alexa's first international launch. Refactored a voice application built English-only to work in any language.
- Migrated the data source powering Alexa's video use-cases to a new format with zero customer-facing regressions.
- Built a personalized TV listings voice experience working closely with Machine Learning Scientists.
- Progressed from SDE I through SDE II to Senior SDE with over 4 years at Sr SDE level.

EXPERIENCE, CONTINUED

eBay.com

May 2012 - Dec 2015

Software Engineer I, Cloud Engineering

Aug 2014 - Dec 2015

- Built and owned Cloud Management Automation for one of the world's top-5 largest OpenStack deployments, responsible for bug fixes, maintenance, and designing new features end-to-end.
- Identified, mitigated and fixed a dangerous bug in eBay's DNS table impacting hundreds of thousands of production IP addresses. Manually fixed in production without negative customer impact.

Software Engineering Intern II, Cloud Engineering

May 2013 - Aug 2013

- Return intern in the Cloud Engineering group embedded with the Cloud Management team, identified and rebuilt Fault-Domain Calculation Component.

Software Engineering Intern I, Zoom Image Storage Platform

May 2012 - Aug 2012

- Built MongoDB Operations Log Replication solution for in-house Image Metadata storage platform.

Data Mining Research Group, UIUC

Aug 2010 - May 2012

Undergraduate Research Assistant

- Coauthored three papers in High-Impact ACM Journals while working as an Undergraduate Research Assistant in Prof. Jiawei Han's Data Mining Research Group at the University of Illinois - Urbana Champaign (UIUC).
- Authored a Senior Thesis titled "Extraordinary Popular Delusions and the Madness of Reddit: The Effect of Anonymous Single Vote Manipulation on In Vivo Social News Networks" based on a 3 month experiment proving the effectiveness of manipulation on Reddit posts via random voting from a single account.

SIDE PROJECTS

Rankem: Built a production-ready social ranking application by running AI agents through a complete development lifecycle: product definition, high-level design, roadmap, and 200+ human-reviewed code changes. Delivered in weeks what was scoped as a multi-quarter team effort.

Tilted: Designed a novel asynchronous poker format (ten concurrent hands, shared stack) and shipped it as an iOS app. Iterated through 5 beta releases against thousands of logged hands of real play; in TestFlight, App Store release in progress.

Sous: Voice-first iOS app that captures what you cooked by dictation and structures into editable and maintainable recipes using on-device language models.

Buster: Built an agent for couples to help decide what movies to watch at night, preventing hours of scrolling with a novel approach to movie recommendations.

AWARDS

Spot Award eBay · 2015

Peer-nominated, paid award for identifying and fixing a systemic issue in DNS capacity management during the eBay/Paypal split.

Innovator Award Family Giving Tree · 2015

Built a donation logistics iPhone app to help the charity coordinate distribution center pickups with volunteer drivers.

PUBLICATIONS

Random Voting Effects in Social-Digital Spaces: A Case Study of Reddit Post Submissions

ACM Hypertext 2015 · Glenski, M., Johnston, T.J., & Weninger, T.

The Parallel Path Framework for Entity Discovery on the Web

ACM TWeb 2013 · Weninger, T., Johnston, T.J., & Han, J.

WINACS: Construction and Analysis of Web-Based CS Information Networks

SIGMOD 2011 · Weninger, T., Danilevsky, M., Fumarola, F., Hailpern, J., Han, J., Ji, M., Johnston, T.J., Kallumadi, S., Kim, H., Li, Z., McCloskey, D., Sun, Y., TeGrotenhuis, N., Wang, C., & Yu, X.